Graphics Programming

Matrices Recap

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Scaling Transformations

- Multiples each component of a vector by a constant
  - $\mathrm{F}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} r \bullet x \\ s \bullet y \end{bmatrix} = \begin{bmatrix} r & 0 \\ 0 & s \end{bmatrix} \bullet \begin{bmatrix} x \\ y \end{bmatrix}$
  - F(): a scaling function that scales x-axis by *r*, and scales y-axis by *s*
    - E.g.,
      - Given a point at [10, 5], assume r=1.2, s=0.6, the new location of the point will be [12, 3]
      - $\mathrm{F}\left(\begin{bmatrix} 10 \\ 5 \end{bmatrix}\right) = \begin{bmatrix} 1.2 \bullet 10 \\ 0.6 \bullet 5 \end{bmatrix} = \begin{bmatrix} 1.2 & 0 \\ 0 & 0.6 \end{bmatrix} \bullet \begin{bmatrix} 10 \\ 5 \end{bmatrix} = \begin{bmatrix} 12 \\ 3 \end{bmatrix}$
- A scaling matrix only has values on the diagonal line

Rotation in 3D

+Around z-axis $\begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$

+Around x-axis $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}$

+Around y-axis $\begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & cos(\theta) \end{bmatrix}$

Homogeneous Coordinates for 3D Translation

$$\begin{bmatrix} 1 & 0 & 0 & m \\ 0 & 1 & 0 & n \\ 0 & 0 & 1 & p \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x + m \\ y + n \\ z + p \\ 1 \end{bmatrix}$$

Final Project Matrix

$$+\begin{bmatrix} \frac{1}{r \bullet \tan(\frac{a}{2})} & 0 & 0 & 0 \\ 0 & \frac{1}{\tan(\frac{a}{2})} & 0 & 0 \\ 0 & 0 & \frac{n+f}{n-f} & \frac{2 \bullet n \bullet f}{n-f} \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

n: near distance f: far distance a: angle of view r: aspect ratio

Global vs Local Transformation

- Apply ***T*** as a **global** transformation
  - Let the new model matrix equal ***T•M***
    - Multiply on LEFT

- Apply ***T*** as a **local** transformation
  - Let the new model matrix equal ***M•T***
    - Multiply on RIGHT